

Definition of swept angle

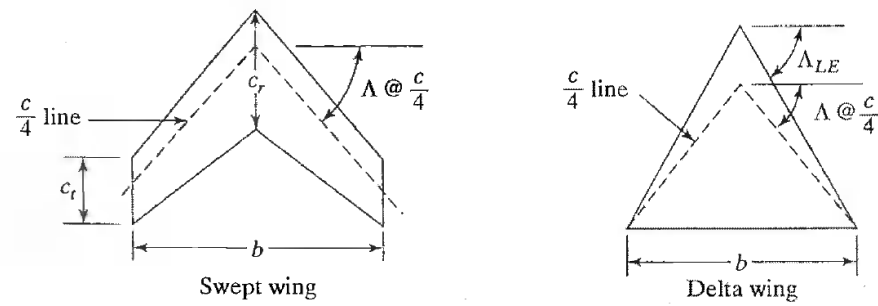


Figure 5.6 Geometric characteristics of the wing planform.

Representation of A_n terms in circulation:

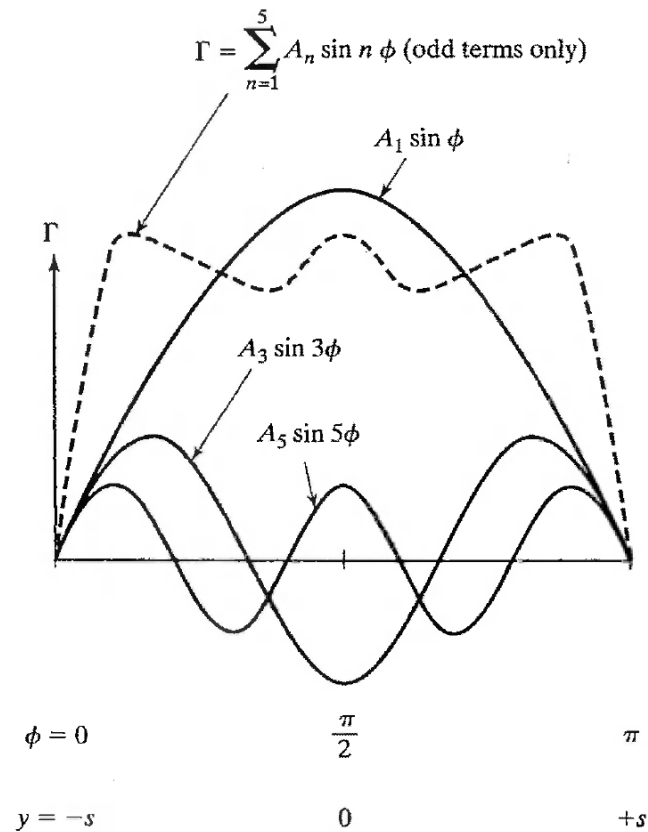
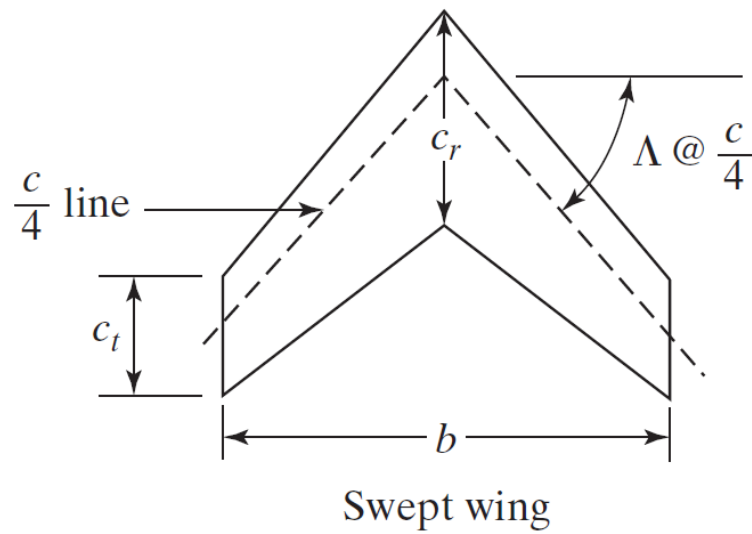


Figure 7.11 Symmetric spanwise lift distribution as represented by a sine series.



Area of a wing with unequal C_r and C_t

$$S = 0.5(c_r + c_t)b = 0.5c_r(1 + \lambda)b$$

Aspect ratio of that wing

$$AR = \frac{2b}{c_r + c_t}$$

This is the equation of lifting line theory for a general distribution, α is the geometric angle of attack and should be given in the problem (usually a constant value for all parts of the wing), but this equation has different values at various locations considering the chord length, $c(\theta_0)$ and $\alpha_{L=0}$ can be different, i.e., the wing consists of different airfoils. Even if they are the same, $\sin \theta_0$ is different depending on the location of θ_0 .

$$\alpha(\theta_0) = \frac{2b}{\pi c(\theta_0)} \sum_1^N A_n \sin n\theta_0 + \alpha_{L=0}(\theta_0) + \sum_1^N n A_n \frac{\sin n\theta_0}{\sin \theta_0}$$